

DSC 335, Handout 11: Simple Normal Regression (*Chapter 9*)

LEARNING OBJECTIVES.

- *Build a Bayesian regression model and simulate its posterior.*
- *Conduct inference using Bayesian regression models.*

Simple Normal Linear Regression

Suppose we have a response variable of interest Y and a predictor X that have a linear relationship. We have n pairs of observations of these variables. One way to model the relationship between these variables is with the Normal simple linear regression model:

The appropriateness of the Normal regression model depends on the following assumptions:

Example 1. We are interested in modeling the relationship between the number of absences a student reports for the previous semester and their college GPA. We will use the data from our class project. Using frequentist methods, estimate the least squares regression line, calculate a 90% confidence interval for the true slope of the linear relationship between the variables, and calculate a 90% prediction interval for a new student with 4 absences.

But wait! We want a Bayesian regression model. Which means we must put priors on any unknown parameter in our model. Tell whether each of the following is a data variable or a parameter, and give the general support of any parameter:

- Y_i
- β_0
- β_1
- σ
- μ_i
- X_i

Bayesian Simple Linear Regression Model

Example 2. Consider the following prior information: We believe that students with a typical number of absences will have a GPA of around 3.5, but that it could be as high as 3.8 or as low as 3.2. We think that each absence causes a decrease of around 0.2 GPA points but that this value could be between 0.02 and 0.38. Finally, we think that the actual GPAs of students with a fixed number of absences will vary with a standard deviation of around 0.5 GPA points. Give reasonable prior distributions for each of the model parameters.

Simulation using `rstanarm`

The `rstanarm` package tailors the power of `rstan` to applied regression models (arm). It contains built in functionality that allow us to skip the model specification (Step 1). For the Normal Bayesian linear model we will use the `stan_glm` function:

Example 3. Use `rstanarm` to simulate the posterior distribution for the parameters in our regression model. Give the equation of the estimated regression line and a 90% credible interval for the slope parameter. How different are these from the frequentist results?

Example 4. Do our results provide convincing evidence that there is a negative linear relationship between absences and college GPA? Explain

Posterior Prediction

As we saw in Chapter 8, the posterior predictive distribution takes into account two types of variability. For a regression model, these are:

Example 5. Calculate and interpret a 95% posterior predictive interval for the GPA of a new student with 4 absences.

Example 6. Rows 27 to 79 of the `c11` data file correspond to spring 2024. Rows 1-26 correspond to fall 2024. The rest correspond to this semester. Simulate a posterior distribution of our parameters sequentially by semester and discuss how our knowledge about the slope has evolved over time.

Default `rstanarm` priors

Example 7. The `mtcars` data frame gives information about a set of cars compiled by Motor Trend magazine in 1974. We want to create a Bayesian model of horsepower (`hp`) as a function of the weight of a car in thousands of pounds (`wt`). Using default priors, estimate the regression line, calculate and interpret a 95% credible interval for the slope, and give a point estimate and 95% posterior predictive interval for the horsepower of a car that weighs 3000 pounds.